

Appendix to v15: Multi-LOO Stress and σ_8 Joint Analysis of LockB and LockC

LVC v15 — Appendix

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Abstract

This appendix to the v15 working paper reports the outcome of two stress tests applied to the two surviving candidates LockB ($z_c = 1/\sqrt{\pi}$, $w = 2/(5\pi)$) and LockC ($z_c = 4/7$, $w = 2/(5\pi)$) in a fully reproduced v14 / v15 environment (Pantheon+ unbinned with full STAT+SYS covariance, Planck 2018 compressed distance priors, SH0ES H_0 , and either DESI DR2 or DR1 BAO). The environment reproduces the v14 Task 2 numbers ($\Delta\text{BIC} = -11.38$ for C1) and the v15 headline LockB/LockC values to within 0.005 in ΔBIC , and the v15 §2 burn of C1 on DR1 ($\Delta\text{BIC} = +0.61$). On this reproduced environment we run a multi-point leave-one-out (K–A) sweep over 22 BAO subset combinations (11 on each release) and a joint analysis with the 19-point σ_8 compilation used in v11 §3, with linear growth solved on the locked $H(z)$. The K–A sweep reveals that the ΔBIC support of LockC concentrates on a single BAO point in each release (LRG2 at $z = 0.706$ for DR2; LRG1 at $z = 0.510$ for DR1); when both are removed simultaneously, ΔBIC moves to +0.02 (DR2) and +4.94 (DR1). All 22 single-point removals match the v15 §5 single-LOO outcomes. The σ_8 joint analysis returns $\chi^2(\sigma_8)$ values for both locks within 0.27 of LCDM and joint ΔBIC values of $-5.50 / -3.86$ (LockB on DR2/DR1) and $-6.37 / -3.82$ (LockC on DR2/DR1), in each case slightly stronger than the geometry-only values. We do not interpret these results; we record them. The characterization in v15 §6 of LockB and LockC as *live*, *not closed* is unchanged.

1 Environment validation

All numerical results in this appendix use the same likelihood as v14 Task 2 generalized to admit an arbitrary locked Gaussian $T(z) = 1 + A \cdot \exp[-((z - z_c)/w)^2]$ with the standing v14 convention that $T(z)$ multiplies the dark-energy term in the high- z sound-horizon integral and multiplies $H(z)$ directly in the late-time geometric integrals. The Pantheon+ likelihood uses the official 1701-SN file Pantheon+SH0ES.dat with the full STAT+SYS covariance (1701×1701) from the 4_DISTANCES_AND_COVAR release, with M_B profiled analytically. The Planck 2018 compressed distance priors (R, l_A, ω_b) are taken from Chen, Huang & Wang (2019), with the multiplicative calibration of v14 applied identically to LCDM and the locked candidates.

Quantity	this appendix	v14 / v15 paper	difference
LCDM Task 2 χ^2	1606.936	1606.94	0.004
C1 Task 2 χ^2 (DR2)	1588.103	1588.10	0.003
C1 ΔBIC (DR2)	-11.379	-11.38	0.001
LCDM χ^2 (DR1)	1598.281	1598.28	0.001
C1 χ^2 (DR1)	1591.439	1591.44	0.001

C1 Δ BIC (DR1, burn)	+0.611	+0.61	0.001
LockB Δ BIC (DR2)	−5.294	−5.29	0.004
LockB Δ BIC (DR1)	−3.603	−3.60	0.003
LockC Δ BIC (DR2)	−6.209	−6.21	0.001
LockC Δ BIC (DR1)	−3.605	−3.60	0.005

Table 1. Reproduction of v14 Task 2 and v15 reference results in the present environment. All target values are reproduced to within 0.005 in Δ BIC. The C1 burn at DR1 is reproduced as +0.611 against the v15 §2 value of +0.61.

2 K–A multi-point leave-one-out for LockC

v15 §5 reports leave-one-out (LOO) over single BAO labels: 12 of 12 BIC pass on DR2 and 10 of 11 BIC pass on DR1, with the single DR1 fail occurring at LRG1 ($z = 0.510$). We extend this to subset removal of pairs, triples, and one quadruple, fixing the lock at $z_c = 4/7$, $w = 2/(5\pi)$ (LockC) and refitting (H_0 , Ω_m , ω_b , A). Eleven distinct subsets are run on each BAO release, drawn from {LRG1, LRG2, BOSS, eBOSS_LRG}. Each subset replaces the single-LOO of v15 §5 with simultaneous removal of multiple labels. The data subsets, Δ BIC values, and verdicts (WIN if Δ BIC < −2; tie if $−2 \leq \Delta$ BIC < +2; BURN if $\geq +2$) are tabulated below.

2.1 DR2 BAO release

Subset removed	dN	Δ BIC	verdict
LRG1 + eBOSS_LRG	4	−6.889	WIN
LRG1	2	−6.306	WIN
eBOSS_LRG	2	−6.113	WIN
LRG1 + BOSS + eBOSS_LRG	6	−5.694	WIN
LRG1 + BOSS	4	−5.102	WIN
BOSS	2	−5.042	WIN
LRG2	2	−0.746	tie
LRG1 + LRG2 + eBOSS_LRG	6	−0.363	tie
LRG1 + LRG2	4	+0.022	tie
LRG1 + LRG2 + BOSS + eBOSS_LRG	8	+0.028	tie
LRG1 + LRG2 + BOSS	6	+0.496	tie

Table 2a. LockC on DR2 BAO with subset removal of the labels {LRG1, LRG2, BOSS, eBOSS_LRG}. Eleven distinct subsets, sorted by Δ BIC. WIN/tie boundary at Δ BIC = −2. Six subsets yield WIN and five yield tie. Among the eleven subsets, the five that include LRG2 cluster in the tie band [−0.75, +0.50]; the six that exclude LRG2 cluster in [−6.89, −5.04].

2.2 DR1 BAO release

Subset removed	dN	Δ BIC	verdict
eBOSS_LRG	2	−3.688	WIN
BOSS	2	−3.168	WIN
LRG2	2	−0.322	tie
LRG1	2	+1.873	tie
LRG1 + eBOSS_LRG	4	+1.892	tie

LRG1 + BOSS + eBOSS_LRG	6	+1.945	tie
LRG1 + BOSS	4	+2.051	BURN
LRG1 + LRG2 + BOSS	6	+4.903	BURN
LRG1 + LRG2	4	+4.942	BURN
LRG1 + LRG2 + BOSS + eBOSS_LRG	8	+5.235	BURN
LRG1 + LRG2 + eBOSS_LRG	6	+5.364	BURN

Table 2b. *LockC on DR1 BAO. Two subsets yield WIN, four tie, and five BURN. The single-LOO entries (rows with $dN = 2$) are LRG1 +1.873 (matching the v15 §5 LOO fail at +1.87), LRG2 -0.322, BOSS -3.168, eBOSS_LRG -3.688. Among the eleven subsets, all eight that include LRG1 lie in the tie/BURN band; the three that exclude LRG1 yield two WIN and one tie. The four BURN entries all include the pair LRG1+LRG2.*

2.3 Subset patterns

Two patterns appear in Tables 2a-2b. On DR2, the ΔBIC support for LockC is concentrated in subsets that retain LRG2 ($z = 0.706$); five of five LRG2-inclusive removals lie in $[-0.75, +0.50]$ and six of six LRG2-exclusive removals lie in $[-6.89, -5.04]$. On DR1, the analogous concentration is in LRG1 ($z = 0.510$); eight of eight LRG1-inclusive removals lie in $[+1.87, +5.36]$ and three of three LRG1-exclusive removals lie in $[-3.69, -0.32]$. The simultaneous removal of LRG1 and LRG2 produces $\Delta\text{BIC} = +0.022$ (DR2) and $+4.942$ (DR1). The single-LOO statements of v15 §5 are recovered in every $dN = 2$ row of these tables, including the only DR1 single-LOO fail at LRG1 (this work: +1.873; v15: +1.87). LockB exhibits the same pattern with magnitudes shifted by the $\Delta\chi^2 \approx 0$ between LockB and LockC on DR1 and $\approx +0.92$ between them on DR2 noted in v15 §5; we do not tabulate the LockB sweep separately.

3 Joint analysis with $f\sigma_8$

We add the 19-point $f\sigma_8$ compilation of v11 §3 to the v14/v15 likelihood, with linear growth $D(z)$ and growth rate $f(z)$ obtained by integrating the standard scale-independent linear-growth ODE with $H(z) = H_{\text{LCDM}}(z) \cdot T(z)$ and initial conditions $D = a$, $dD/d\ln a = a$ at $a_{\text{init}} = 10^{-3}$. The amplitude $\sigma_8(0)$ is profiled analytically. As an environment check the present integrator returns $\chi^2(f\sigma_8) = 11.99$ at LCDM with $(\Omega_m = 0.315, \sigma_8 = 0.811)$, matching the v11 §3.2 sanity value 11.99 to two decimals; at $(\Omega_m = 0.227, \sigma_8 = 0.845)$ it returns $\chi^2 = 9.21$, matching the v11 RSD-best 9.21. The background parameters for each lock fit are taken from the joint geometric+SH0ES+DP best fit of §1 above; only $\sigma_8(0)$ is varied in the $f\sigma_8$ evaluation.

3.1 $f\sigma_8$ alone

Model	BAO release	Ω_m	$\sigma_8(0)$	$\chi^2(f\sigma_8)$	$\Delta\chi^2$ vs LCDM
LCDM	DR2	0.2775	0.8138	10.105	–
LCDM	DR1	0.2812	0.8119	10.227	–
LockB	DR2	0.2846	0.8219	9.890	–0.215
LockB	DR1	0.2867	0.8225	9.963	–0.264
LockC	DR2	0.2848	0.8224	9.934	–0.171
LockC	DR1	0.2867	0.8229	10.004	–0.223

Table 3. $f\sigma_8$ -only fit at each model's joint best-fit Ω_m , with $\sigma_8(0)$ profiled. All four lock entries lie within 0.27 of LCDM in $\chi^2(f\sigma_8)$; all four are negative, indicating the $f\sigma_8$ data are fitted slightly better at the lock-preferred Ω_m (≈ 0.285) than at the LCDM-preferred Ω_m (≈ 0.278) on the present 19-point compilation. Both lock and LCDM χ^2/N values are below 0.6, characteristic of the compilation.

3.2 Geometry + $f\sigma_8$ joint

Model	BAO	$\chi^2(\text{geom})$	$\chi^2(f\sigma_8)$	Total χ^2	ΔBIC vs LCDM
LCDM	DR2	1606.936	10.105	1617.041	baseline
LCDM	DR1	1598.281	10.227	1608.508	baseline
LockB	DR2	1594.188	9.890	1604.078	–5.50
LockB	DR1	1587.225	9.963	1597.188	–3.86
LockC	DR2	1593.273	9.934	1603.207	–6.37
LockC	DR1	1587.223	10.004	1597.227	–3.82

Table 4. Joint geometry + $f\sigma_8$ analysis. $N = 1746$ (DR2) or 1744 (DR1). Both locks add one parameter (A) over LCDM in the joint count ($k = 5$ vs $k = 4$). Joint ΔBIC values are shifted from the geometry-only values of v15 §5 (LockB DR2 –5.29, LockB DR1 –3.60, LockC DR2 –6.21, LockC DR1 –3.60) by –0.21, –0.26, –0.16, –0.21 respectively. The shifts are negative in all four cases.

4 Status

The present appendix records three numerical findings on the LockB and LockC candidates of v15 §5, in the v14/v15 environment reproduced to within 0.005 in ΔBIC of all v14 Task 2 and v15 headline values:

- (i) The single-LOO results of v15 §5 are reproduced verbatim in the $dN = 2$ rows of Tables 2a-2b; in particular the unique DR1 single-LOO fail at LRG1 is recovered as $\Delta\text{BIC} =$

+1.873.

- (ii) Multi-point subset removal exhibits a release-specific concentration: on DR2 the LockC ΔBIC band $[-6.89, -5.04]$ is realized only in subsets that exclude LRG2 ($z = 0.706$), and the band $[-0.75, +0.50]$ is realized only in subsets that include LRG2; on DR1 the corresponding pivot point is LRG1 ($z = 0.510$), with simultaneous removal of LRG1+LRG2 yielding $\Delta\text{BIC} = +0.022$ on DR2 and $+4.942$ on DR1.
- (iii) Adding the 19-point $f\sigma_8$ compilation reduces $\chi^2(f\sigma_8)$ by $0.17\text{--}0.27$ for both locks relative to ΛCDM ; the joint ΔBIC values ($-5.50/-3.86$ for LockB, $-6.37/-3.82$ for LockC) are shifted negatively by $0.16\text{--}0.26$ from the v15 geometry-only values.

These results are recorded without interpretation. The v15 §6 characterization of LockB and LockC as *live, not closed* is unchanged by this appendix.

5 Reproducibility

All numerical values in this appendix are produced by an extension of the v14 reproduce script (`lvc_v14_reproduce.py`) that admits an arbitrary locked Gaussian (z_c, w) in the joint ($H_0, \Omega_m, \omega_b, A$) fit, supports DR1 BAO as an alternative to DR2, supports the K–A subset removal sweep, and integrates the linear growth ODE for arbitrary $T(z)$. The Pantheon+ data and covariance (Pantheon+SH0ES.dat, Pantheon+SH0ES_STAT+SYS.cov) are obtained verbatim from github.com/PantheonPlusSH0ES/DataRelease, subdirectory Pantheon%2B_Data/4_DISTANCES_AND_COVAR. The Planck distance priors and their calibration are those of v14. The 19-point $f\sigma_8$ compilation is that of v11 §3. Tabulated χ^2 values reproduce to within 0.05; tabulated ΔBIC values reproduce to within 0.10. Total runtime to reproduce all tables in this appendix is approximately 90 minutes on a single CPU core, dominated by the K–A sweep (22 fits at ≈ 90 s each) and the joint $f\sigma_8$ evaluations.

Note on scope

This appendix supplements but does not supersede the v15 working paper. It records the outcome of two stress tests applied to the two surviving candidates of v15 §5. It introduces no new candidate, makes no statement about candidates not present in v15, and does not advance the question raised in v15 §6 of whether either of LockB or LockC closes as a natural-constant ansatz across both BAO releases.